



FCTUC FACULDADE DE CIÊNCIAS
E TECNOLOGIA
UNIVERSIDADE DE COIMBRA

Business Intelligence 2024-25
DEI, FCT, University of Coimbra

BI Project

Data Warehouse, OLAP and Data Mining

Deliverables and deadlines

1. **Project idea:** 21st of February 2025
2. **Part 1:** 14th of March 2025
 - a. Submit all the files used for the development of this part, e.g., report, code, data to *Inforestudante*.
 - b. Submit a .pdf file to *Inforestudante* with your presentation slides.
 - c. Deliver a 10-minute presentation in next week's classes, make a demonstration of your project
3. **Part 2:** 16th of May 2024
 - a. Submit all the files used for the development of this part, e.g., report, code, data to *Inforestudante*.
 - b. Submit a .pdf file to *Inforestudante* with your presentation slides.
 - c. Deliver a 10-minute presentation in next week's classes, make a demonstration of your project

Project Idea

Ralph Kimball identified three main success factors for a BI project:

1. The level of commitment and sponsorship of the project from senior management
2. The level of the business need for creating a BI implementation
3. The amount and quality of business data available

To begin this project, you must **start by gaining support from the senior management** (in this case, the instructors) for your project. Thus, you must convince management of the importance and relevance of your BI project. **To do so, you must submit a document explaining your project idea. The subject you select must have data that allows**

you to implement both OLAP and Data Mining approaches.

You must identify the team and specific objectives of your project.

List of relevant information you should provide:

- 1. Identify the team of 3 students.**
- 2. Identify the title of the project.**
- 3. Describe the context for your work:**
 - a. Present the customer (the one paying for your work)
 - b. Present the "business":
 - i. *Products or services, location, clients, etc.*
 - c. Identify the Key Performance Indicators (KPI) of the business
 - i. *In your own words, describe what influences these KPI*
 - d. Identify the main challenges that the company faces to grow
 - i. *You don't need to be very thorough! Use your insight and shared knowledge to discuss: Competitors, new products, new tendencies, new markets, distribution/transportation issues, cheaper services, royalties, human resources, etc.*
- 4. Describe your data source**
 - a. Identify the location of the data that you will use in the project
 - b. Use a dataset that you have access to and a license to use, e.g., from
<http://www.kdnuggets.com/datasets/index.html>
<http://archive.ics.uci.edu/ml/>
<https://www.rdatamining.com/datasets>
<https://www.kaggle.com/datasets>
<https://datasetsearch.research.google.com/>
<https://opendata.cityofnewyork.us/data/>
 - c. Explain (gross grain) how you will access the data
 - d. Describe the amount and the quality of data that you will have at your disposal
 - i. *Estimate how many years of data can be accessed, how many "records," how many megabytes (if you can), etc.*
 - ii. *Estimate how often the data should be refreshed to get the most up-to-date results*
 - e. Describe potential issues with the quality of data
 - i. *Wrong data input by the system users, lack of compatibility between data in different locations, missing information, etc.*
- 5. Explain the objective of your BI solution**
 - a. Explain how your BI solution can help the business thrive
 - b. Identify the business processes for which you can provide new and relevant information based on the data available
 - c. Explain why this information is useful
 - i. How can this information be used to support business decisions
 - ii. Who in the customer organization would be interested in seeing your results

- d. Identify a set of the most relevant questions you believe you can find answers to in your data
 - i. E.g., "What are our best-selling products?", "What is the average age of our best clients?", "What has been our profit in the last five years?" etc.
 - ii. Clearly state what problems would be handled using Machine Learning approaches, such as prediction, classification, etc.
 - iii. These questions are not definitive, they can change, but they are helpful to understand better the problem that you are proposing to solve
 - iv. In the final presentation, there should be a much larger set of questions and answers

Part 1 – Data Warehousing and OLAP

1.1. Requirements

Now it is time to define the **requirements** for your project (the BIG questions). And, since this project is all about data, we will focus on a subset of the full range of requirements usually associated with a software project. We will leave out technological aspects, non-functional requirements (quality attributes), and how the overall system should function.

In this stage, we will focus on how data can be combined and exhibited on your BI solution: **User Interface and Reports**. Thus, this is not a complete requirements elicitation process, which would take much more time than we have available. It is a process specifically tailored to the needs of this project.

Your objective for this phase is to understand the raw data you have available to work with and decide the best way to expose the information you will extract from this data to non-technical users. Thus, you convince management that you will deliver a BI solution that fits the project's needs. **Be sure to include drill-down and roll-up actions and lots of slice-and-dice capabilities.**

Important Notes

This phase is one of the most important ones you will have in the project because it will give you a target, an objective for which you will be working. But the result of this phase will not be complete because:

1. You don't know the data very well.
2. You don't know everything you can do with the data (e.g., data mining, predictive analytics).
3. You have no experience in the BI field.

On the other hand, the results from this iteration will represent the more significant part of the objectives of this work. Thus, it must be as complete and correct as possible. Keep in mind that the data you will be using is still raw and that it will be your job to prepare, clean, and transform this data into a form that might suit your interests better.

To reach/refine the requirements you should also create mockups for the screens and documents you will be provided with your BI solution. If for the documents you can simply use MS Word or something similar, for the screens, it is much simpler to use a mockup tool such as <https://gomockingbird.com> or <https://balsamiq.com/> (or even tableau). Use the mockup tool to describe how users interact with the data and how the information will be displayed.

The objective is the selection of the tools to use for storing data, ETL, and OLAP. Thus, it is essential to:

1. Identify the "must-have" requirements for your tools. For instance:
 - a. What features should the ETL tool have? Visual modeling, process flow definition, etc.
 - b. What is the size of the DW for the server to handle?
 - c. Should the OLAP tool be web-based?
 - d. Freeware? Open source? Can it be a trial? Most used? Most recent?
 - e. Among others.
2. Describe the criteria for comparing the different solutions based on the elicited requirements.
3. Describe the approach for searching and comparing the multiple software solutions.
4. Present the results and identify the list of software that you will be using.

1.4. ETL and OLAP

With the data model and the DW software selected, it is time to get the "job" done. In this phase, you will focus on planning the ETL process, executing the ETL, and creating the tools for your user OLAP activities.

You must explain your ETL process and your OLAP solutions. The objective is to present the complete ETL plan for the first and all the subsequent loads of the DW and the OLAP/Reporting tools you developed. Thus, it is essential to:

1. Identify and describe the sources of data.
2. Present the overall ETL plan for your solution.
3. Describe the staging area.
4. For each significant action in the plan, explain why it is necessary and how it is implemented.
5. Present the major challenges to the implementation.
6. Present the following metrics: the size of the source data, the size of data on the 1st load, the size of data on the subsequent loads, the time elapsed on the first load, time used for each update.
7. Identify problems with the source and DW data that were not dealt with (if they exist).
8. Explain how the ETL process is automated to allow future data updates in the DW and the update strategy for the dimensions, facts tables, and frequency.
9. Describe how the OLAP data is presented to final users, how they can access it, and modify the search parameters.
10. Describe the analyses being performed.
11. Present and discuss the initial findings.
12. Discuss results from the business perspective and explain how the information can be used for Decision Support.

Explain the strategies and techniques for optimizing the DW and the OLAP queries performance (views, indexes, partitioning, etc.). **Next, it is time to build the front end of the project.** Now that you have all the data ready in the DW, using the mockups as guidelines, make the user interfaces for your BI application using the software you selected for this effect. Create the necessary UI to provide your users with the means to answer their questions with the data you prepared. Make sure to include exciting visualizations, such as pivot tables, graphs, etc. And have all the necessary controls for allowing the "slice and dice" of information, the "drill-down" and "roll-up" of the data, etc.

Part 2 – Data Mining

Data mining tools complement OLAP data analyses with descriptive and predictive capabilities. In this assignment, you will prepare your dataset and use it to explore two knowledge extraction techniques of your choice. But one of the two choices must be a classification study and the other a different study like clustering, association, time series analysis, etc.

This phase aims to describe the data mining approach you used and show how it is integrated into your product and made available to end users. Thus, it is essential to:

1. Describe the main goals of the Data mining project, and do not forget to detail the two proposed techniques.
 - a. For each technique, detail the specific goals
 - b. Justify the algorithm choice for each analysis
2. Identify the software being used.
3. Present the data source and identify the attributes being used.
4. Describe each step of the data mining process:
 - a. Get data
 - b. Clean, prepare and manipulate data, e.g., feature Selection and Reduction
 - c. Detail the design of experiences, namely the hyperparameter search/choice
 - d. Train and test the models
 - e. Test model
 - f. Improve
5. Discuss the results of the study. Present and discuss the results obtained in your data mining assignment. Compare your results with the results from other sources. In this problem, one important aspect is to evaluate among the data available the more appropriate for the different scenarios.
6. Describe how your models are made available to end users and integrated into your BI product.

Final Note on AI

The use of AI tools and technologies, including Generative AI tools (e.g., ChatGPT, Claude, Gemini, or similar LLMs), to assist with coding and writing is permitted. However, to maintain transparency, authors must fully and honestly disclose any utilization of AI-assisted tools in their work, particularly generative AI technologies (e.g., ChatGPT, Claude, Gemini, or similar LLMs). Disclosures should encompass both writing and coding contributions. For instance, authors might include statements in the Acknowledgements/Disclaiming section at the end of the paper such as: “ChatGPT was used to generate a draft of Section X,” or “Claude was utilized to polish English of paragraph Y on page Z.” Additionally, statements like “Gemini was employed to produce tables, graphs, code, data, citations, etc.” can provide clarity on the specific role of AI tools.